Machine Learning A (2026)
Home Assignment 7

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1 Conceptual Questions

1. **False**: but in expectation it is true. That is for the output constructed using k-means++, the corresponding cost function satifies[1]

$$\mathbb{E}\left[\phi\right] \leq 8(\log k + 2)\phi^{\star},$$

   where $\phi^{\star}$ is the lowest possible k-means cluster cost.

2. k-means clustering and PCA are both *unsupervised learning* methods and thus somehow try to extract patterns for understanding and or visualisation of the data. Typically in a data-exploratory state.

2 k-means vs. k-means++

2.1 Part 1: k-means++ theory

Generally for k-means++ the probability that $x \in \mathcal{X}$ is chosen as c_i $(i < k)$ when initial clustering centers $S = \{c_{1,2,\ldots,i-1}\}$ are already chosen is

$$\mathbb{P}\left(c_i = x \mid \{c_{1,2,\ldots,i-1}\}\right) = \frac{D(x, \{c_{1,2,\ldots,i-1}\})^2}{\sum_{u \in \mathcal{X}} D(u, \{c_{1,2,\ldots,i-1}\})^2}, \tag{1}$$

where

$$D(x, \{c_{1,2,\ldots,i-1}\}) = \min_{z \in \{c_{1,2,\ldots,i-1}\}} ||x - z||_2$$

Therefore in the *small-dataset* with $c_1 = A = (-2, 2)$ and $c_2 = D = (7, -5)$ already we

[1] Sadegh Talebi, Clustering: k-means and Beyond, slide 33

have (where i only calculate visually ambiguous distances)

$$\begin{aligned} D\left(B,\{A,D\}\right) &= D\left((0,4),\{(-2,2),(7,-5)\}\right) \\ &= ||(0,4)-(-2,2)|| \\ &= \sqrt{8} \end{aligned}$$

$$\begin{aligned} D\left(C,\{A,D\}\right) &= D\left((6,3),\{(-2,2),(7,-5)\}\right) \\ &= \min\left\{||(6,3)-(-2,2)||_2, ||(6,3)-(7,-5)||_2\right\} \\ &= \min\left\{\sqrt{65},\sqrt{65}\right\} \\ &= \sqrt{65} \end{aligned}$$

$$\begin{aligned} D\left(E,\{A,D\}\right) &= D\left((4,-6),\{(-2,2),(7,-5)\}\right) \\ &= ||(4,-6)-(7,-5)||_2 \\ &= \sqrt{10} \end{aligned}$$

$$\begin{aligned} D\left(F,\{A,D\}\right) &= D\left((0,-4),\{(-2,2),(7,-5)\}\right) \\ &= \min\left\{||(0,-4)-(-2,2)||_2, ||(0,-4)-(7,-5)||_2\right\} \\ &= \min\left\{\sqrt{40},\sqrt{50}\right\} \\ &= \sqrt{40}. \end{aligned}$$

From this we simply apply (1) yielding the following probability distribution

$$\begin{aligned} \mathbb{P}\left(c_3 = B\right) &= \frac{8}{123} \\ \mathbb{P}\left(c_3 = C\right) &= \frac{65}{123} \\ \mathbb{P}\left(c_3 = E\right) &= \frac{10}{123} \\ \mathbb{P}\left(c_3 = F\right) &= \frac{40}{123}. \end{aligned}$$

Note that strictly speaking (1) also has $D\left(A,\{A,D\}\right) + D\left(D,\{A,D\}\right) = 0$ in the denominator.

2.2 Part 2: Comparing seeding methods for k-means clustering

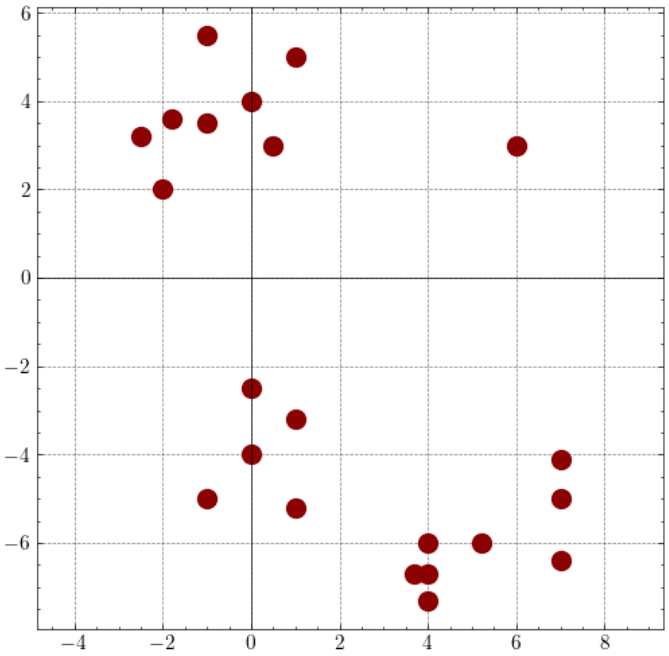


Figure 1: Visualization of “big-dataset”

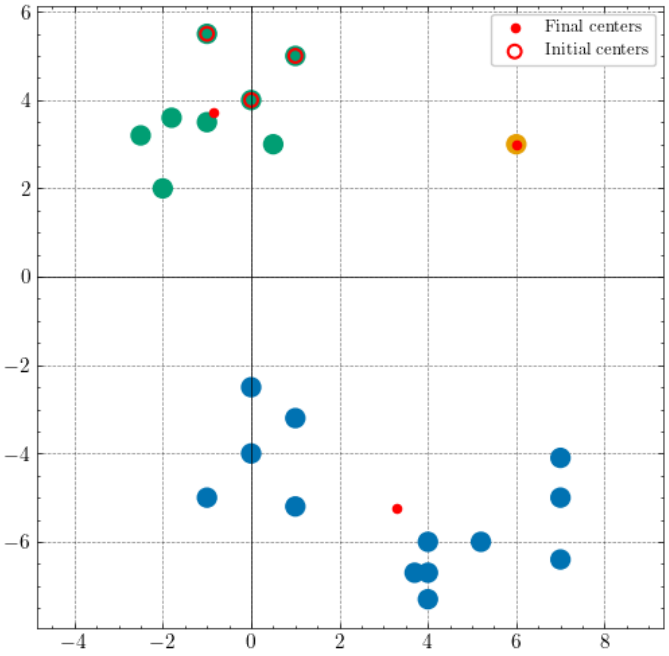


Figure 2: Visualization of “big-dataset”

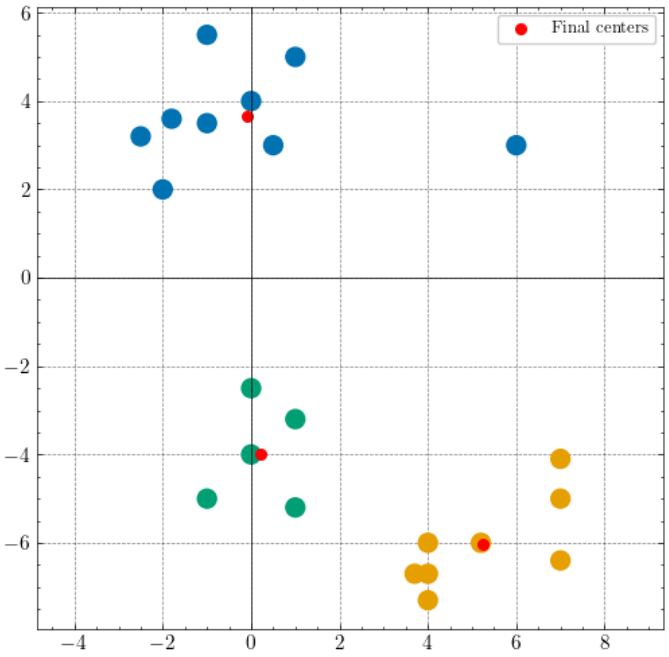


Figure 3: Visualization of “big-dataset”

3 Visualization

4 Determining the Number of Clusters

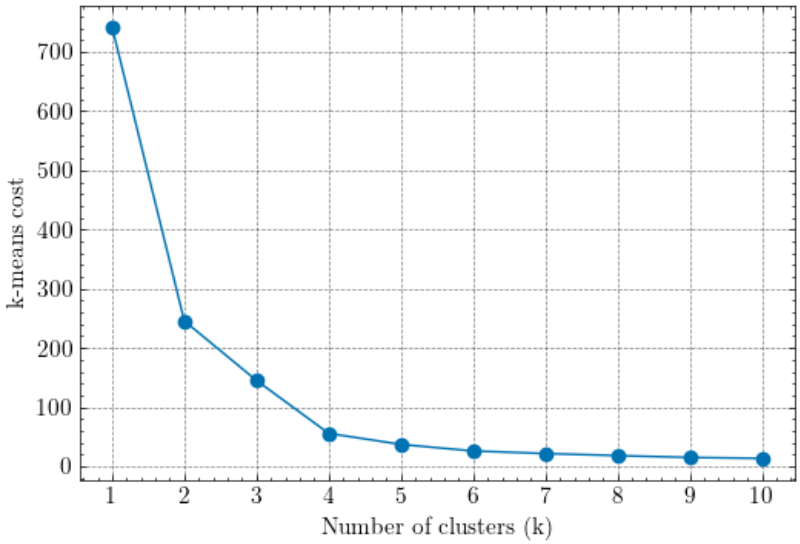


Figure 4: Visualization of “big-dataset”